METRIC

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10 February 2025
SUPERSEDING
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14 October 2010

PERFORMANCE SPECIFICATION

HYDRAULIC FLUID, PETROLEUM-BASED, INHIBITED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers inhibited petroleum-based hydraulic fluids for use in hydraulic systems and in other applications where a high-grade hydraulic fluid having anti-corrosion and anti-oxidation properties is required. This hydraulic fluid is not an extreme-pressure or anti-wear fluid. This hydraulic fluid should not be used in systems where a fire-resistant fluid is required.

1.2 Classification. The hydraulic fluids covered by this specification are of the following International Organization for Standardization (ISO) viscosity grades, as specified (see 6.2):

ISO Viscosity Grade	Military Symbol	Grade Code
32	2075-T-H	A
46	2110-T-H (NATO symbol H-573)	B
68	2135-T-H	C

1.3 Part or identifying number (PIN). PINs to be used for petroleum-based hydraulic fluids acquired to this specification are created as follows:

M	17672	-	X
Prefix for Military Specification	Specification Number	-	Grade (see code in 1.2)
Examples: M17672-A and M17672-C			

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

(Copies of the chapter titled “Material Control Program” are available by email request to CommandStandards@navy.mil.)

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

ASTM D97 - Standard Test Method for Pour Point of Petroleum Products

ASTM D130 - Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test

ASTM D189 - Standard Test Method for Conradson Carbon Residue of Petroleum Products

ASTM D287 - Standard Test Method for API Gravity of Crude Petroleum and Petroleum Products (Hydrometer/Method)

ASTM D445 - Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)

ASTM D524 - Standard Test Method for Ramsbottom Carbon Residue of Petroleum Products

ASTM D664 - Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration

ASTM D665 - Standard Test Method for Rust-Preventing Characteristics of Inhibited Mineral Oil in the Presence of Water

ASTM D874 - Standard Test Method for Sulfated Ash from Lubricating Oils and Additives

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ASTM D892 - Standard Test Method for Foaming Characteristics of Lubricating Oils

ASTM D943 - Standard Test Method for Oxidation Characteristics of Inhibited Mineral Oils

ASTM D974 - Standard Test Method for Acid and Base Number by Color-Indicator Titration

ASTM D1298 - Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method

ASTM D1401 - Standard Test Method for Water Separability of Petroleum Oils and Synthetic Fluids

ASTM D1500 - Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)

ASTM D1552 - Standard Test Method for Sulfur in Petroleum Products by High Temperature Combustion and Infrared (IR) Detection or Thermal Conductivity Detection (TCD)

ASTM D2270 - Standard Practice for Calculating Viscosity Index from Kinematic Viscosity at 40 °C and 100 °C

ASTM D2272 - Standard Test Method for Oxidation Stability of Steam Turbine Oils by Rotating Pressure Vessel

ASTM D2622 - Standard Test Method for Sulfur in Petroleum Products by Wavelength Dispersive X-ray Fluorescence Spectrometry

ASTM D4052 - Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D4177 - Standard Practice for Automatic Sampling of Petroleum and Petroleum Products

ASTM D4310 - Standard Test Method for Determination of Sludging and Corrosion Tendencies of Inhibited Mineral Oils

ASTM D4530 - Standard Test Method for Determination of Carbon Residue (Micro Method)

ASTM D4898 - Standard Test Method for Insoluble Contamination of Hydraulic Fluids by Gravimetric Analysis

ASTM D4927 - Standard Test Methods for Elemental Analysis of Lubricant and Additive Components – Barium, Calcium, Phosphorus, Sulfur, and Zinc by Wavelength-Dispersive X-Ray Fluorescence Spectroscopy

ASTM D5949 - Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)

ASTM D5950 - Standard Test Method for Pour Point of Petroleum Products (Automatic Tilt Method)

ASTM D6304 - Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration

ASTM D6443 - Standard Test Method for Determination of Calcium, Chlorine, Copper, Magnesium, Phosphorus, Sulfur, and Zinc in Unused Lubricating Oils and Additives by Wavelength Dispersive X-ray Fluorescence Spectrometry (Mathematical Correction Procedure)

ASTM D6481 - Standard Test Method for Determination of Phosphorus, Sulfur, Calcium, and Zinc in Lubrication Oils by Energy Dispersive X-ray Fluorescence Spectroscopy

ASTM E29 - Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The hydraulic fluid furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Products List before contract award (see 4.2 and 6.3).

3.2 Material. The hydraulic fluid shall be a blend of virgin petroleum-based hydraulic fluid stocks plus additive agents, if necessary, to meet the requirements of this specification. Additives, if used to meet the pour point requirement, shall not exceed 0.5 percent by weight. Changes to the additive components in the sample granted qualification approval is not authorized without the approval of the qualifying activity.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Compatibility. The hydraulic fluid shall be compatible with the reference hydraulic fluid(s) furnished by the Government when tested as specified in 4.4.2. For information on obtaining reference hydraulic fluid(s), see 6.7.

3.4 Homogeneity. Additive agents, when used, shall remain uniformly distributed throughout the hydraulic fluid at all temperatures above the pour point and up to 130 °C (266 °F). When visually inspected in accordance with 4.4.3, the hydraulic fluid shall not exhibit sedimentation or separation of insoluble materials.

3.5 Appearance. The hydraulic fluid shall be free of haze or cloudiness (see 4.4.4).

3.6 Chemical and physical requirements. The hydraulic fluid shall conform to the chemical and physical requirements specified in table I.

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TABLE I. Chemical and physical requirements and test methods.

Characteristic	Limit			Test Method [1/]
	2075-T-H	2110-T-H	2135-T-H	
American Petroleum Institute (API) Gravity	Report [2/]	Report [2/]	Report [2/]	ASTM D287, ASTM D1298 (R), ASTM D4052
Pour Point, °C (°F) maximum	-29 (-20)	-23 (-10)	-18 (0)	ASTM D97 (R), ASTM D5950, ASTM D5949
Flash Point, °C (°F) minimum	157 (315)	163 (325)	171 (340)	ASTM D92
Viscosity, centistokes (cSt) @ 100 °C (212 °F)	Report [2/]	Report [2/]	Report [2/]	ASTM D445
Viscosity, centistokes (cSt) @ 40 °C (104 °F)	28.8 – 35.2	41.4 – 50.6	61.2 – 74.8	ASTM D445
Viscosity Index, minimum	94	94	94	ASTM D2270
Acid Number, milligram KOH/gram, maximum	0.20	0.20	0.20	ASTM D974, ASTM D664 (R)
Corrosion, copper strip @ 100 °C (212 °F), maximum	1	1	1	ASTM D130
Rust Prevention (in the presence of salt water)	Pass	Pass	Pass	ASTM D665 procedure B [3/]
Water, percent maximum	0.01	0.01	0.01	ASTM D6304
Sulfated Ash, mass percent	Report [2/]	Report [2/]	Report [2/]	ASTM D874
Foam Characteristics, after blowing/after 10 minutes:				ASTM D892 [4/]
Sequence I, milliliter maximum	65/0	65/0	65/0	
Sequence II, milliliter maximum	65/0	65/0	65/0	
Sequence III, milliliter maximum	65/0	65/0	65/0	
Emulsion Test, after 30 minutes settling time:				ASTM D1401 [5/]
Oil Layer/Water Layer/Emulsion Layer, maximum	41/-/3	41/-/3	41/-/3	
Oxidation Test, after 1000 hours:				ASTM D943
Acid Number, milligram KOH/gram, maximum	2.0	2.0	2.0	ASTM D974, ASTM D664 (R)
Total Sludge, milligram maximum	100	100	100	ASTM D4310
Total Iron, milligram maximum	100	100	100	ASTM D4310 [6/]
Total Copper, milligram maximum	100	100	100	ASTM D4310
Oxidation by Rotating Pressure Vessel, minutes to failure, minimum	Report [2/]	Report [2/]	Report [2/]	ASTM D2272
Solid Particle Contamination, milligram/100 milliliter maximum [7/]	2.5	2.5	2.5	ASTM D4898
ASTM Color	Report [2/]	Report [2/]	Report [2/]	ASTM D1500

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TABLE I. Chemical and physical requirements and test methods – Continued.

Characteristic	Limit			Test Method
	2075-T-H	2110-T-H	2135-T-H	
Sulfur, weight percent	Report 2/	Report 2/	Report 2/	ASTM D1552, ASTM D2622 (R), ASTM D4927, ASTM D6443, ASTM D6481
Conradson Carbon Residue	Report 2/	Report 2/	Report 2/	ASTM D189, ASTM D524 8/, ASTM D4530 (R)

NOTES:

1/ When applicable, the referee test method is followed by the notation, "(R)".

2/ No limiting requirements. Test values shall be recorded in the test report.

3/ Prior to the test, the fluid shall be water washed as follows: 300 grams of the fluid to be tested shall be stirred with 50 grams of water for 30 minutes at 90 °C (194 °F) using the beaker and stirrer in accordance with ASTM D665. After 30 minutes of being stirred, the mixture shall be transferred to a separatory funnel and left to separate. Then the water layer shall be drawn off and the fluid submitted to the saltwater corrosion test.

4/ Option A is not allowed on freshly blended oils. If testing is done more than 1 week after blending, Option A is allowed. A ring of bubbles around the edge of the graduated cylinder shall be considered complete collapse or 0 milliliters.

5/ Distilled water and a stirring temperature of 54 °C (130 °F) shall be used. Discrete water droplets entrained in the oil layer and clinging to the wall of the cylinder may be added to the water layer volume at the rate of 0.1 milliliter per droplet. The results shall be recorded using the format of oil layer/water layer/emulsions layer (time in minutes). There is no requirement for water layer. The emulsion limit applies to any of the three types of emulsions defined in the test method. If more than one emulsion type is present, the 3-milliliter maximum limit applies to the combined volume of the different emulsion types.

6/ The total iron content shall be determined using the procedure and calculations described in sections 12 and 13 of ASTM D4310, except that iron shall be substituted in place of copper where appropriate.

7/ The sample shall be tested in accordance with ASTM D4898 using 0.8-micrometer pore diameter membrane filters.

8/ If ASTM D524 is used to determine carbon residue, the recorded result shall be converted from Ramsbottom to Conradson using the correlation data within ASTM D189 or ASTM D524.

3.7 Toxicity and prohibited materials.

3.7.1 Toxicity. When evaluated in accordance with 4.4.5.1, the hydraulic fluid shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.4.5.1 and 6.6).

3.7.2 Prohibited materials. The hydraulic fluid shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.7.2.1 Additives. The use of additives to improve the viscosity-temperature characteristics of the base stocks is not permitted. Additives, if used, shall not contain chlorine or zinc materials.

3.8 Off-gassing. The hydraulic fluid shall be evaluated for off-gassing in accordance with the requirements of 4.4.6. Based on the circumstances of use and the chemical nature of the hydraulic fluid, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the hydraulic fluid shall be assigned to either the "Permitted" or "Limited" category (see 4.4.6 and 6.5).

3.9 2,6-Di-tert-butylphenol (DTBP) prohibition. The addition of DTBP (Chemical Abstract Service Registry Number 128-39-2) in any hydraulic fluid produced under this specification is prohibited; this applies to both surface

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ship and submarine use. DTBP shall not be intentionally added to the fluid. The level of DTBP, if detected, shall not exceed 10 parts per million by mass (see 4.4.7).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall consist of the tests specified in table II to satisfy the requirements in section 3 and table I.

TABLE II. Qualification and conformance inspections.

Characteristic	Requirement	Verification	Qualification	Conformance
Materials	3.2	---	X	---
Compatibility	3.3	4.4.2	X	---
Homogeneity	3.4	4.4.3	X	X
Appearance	3.5	4.4.4	X	X
Chemical and physical requirements	3.6	4.4.8	X	X
Toxicity and prohibited materials	3.7	4.4.5	X	---
Off-gassing	3.8	4.4.6	X	---
DTBP	3.9	4.4.7	X	---

4.2.1 Change approval. Any change in product composition, formulation, ingredients, production processes, or production location used in the manufacturing of hydraulic fluid that has been qualified shall require written approval of NAVSEA. Incorporation of any changes that have not been approved shall require requalification of the item in question. Unless otherwise approved by NAVSEA, requalification shall include the tests specified in table II.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.2.5 and the tests specified in table II, except the 1000-hour oxidation test. Samples shall be selected as specified in 4.3.2 The tests shall be performed on the finished product.

4.3.1 Lot acceptance requirements. The results from each sample selected as specified in 4.3.2, when tested as specified in table I, shall not vary from the analysis of the sample given qualification approval by more than the tolerances shown in table III.

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TABLE III. Tolerances of physical properties.

Property	Tolerance
API Gravity, degrees API	±1.0
Sulfur, percent:	---
Original sulfur level ≥ 0.15	±0.03
Original sulfur level < 0.15	+0.08 allowable
Viscosity @ 40 °C (104 °F), centistokes 1/	±8
Pour Point, °C 1/	+3
Oxidation by Rotating Pressure Vessel, percent	-15

NOTE:
1/ Samples shall meet the requirements specified in table I.

4.3.2 Sampling for conformance.

4.3.2.1 Inspection lot.

4.3.2.1.1 At place of manufacture. For the purpose of sampling, one lot of hydraulic fluid shall consist of a manufacturer's batch. When material cannot be identified by batch, a lot shall consist of not more than 38 cubic meters (10,000 gallons).

4.3.2.1.2 At place of delivery. For the purpose of sampling, one lot of hydraulic fluid shall consist of all fluid in a single shipment.

4.3.2.2 Sampling a storage tank. A representative sample of 19 liters (5 gallons) of hydraulic fluid shall be taken from each lot in accordance with ASTM D4057.

4.3.2.3 Sampling during loading. Samples shall be drawn at the discharge pipe when it enters the bulk carriers (tank cars, tank trucks, and so forth). At least four samples shall be taken at regular intervals during the entire period of loading or filling, each sample being 473 milliliters (1 pint). Only one sample for one continuous drum filling operation is required and shall be taken after approximately 10 percent of the volume of product has been drummed.

4.3.2.4 Sampling of filled containers. Where the hydraulic fluid is contained in drums or cans, 0.95-liter (1-quart) samples shall be drawn from four randomly selected drums in the lot and composited, unless the lot consists of less than five containers, in which case a 3.78-liter (1-gallon) sample shall be drawn from one drum. The contents of each container to be sampled shall be shaken, rolled, or stirred to mix the contents thoroughly. Equal portions shall be withdrawn from approximately the center of each drum by means of a thief or other sampling device.

4.3.2.5 Examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with ASQ/ANSI Z1.4 at inspection level II to verify compliance regarding fill, closure, marking, and other requirements not involving tests. Samples shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each container shall also be weighed to determine quantity. Any container in the sample having one or more defects or under required fill shall be cause for rejection of the container.

4.3.3 Rejection of lots. When a sample of hydraulic fluid fails any of the tests of 4.4, this shall be cause for rejection of the lot represented by the sample.

4.3.4 Sampling for retainment. A composite sample from the blending tank sufficient for testing with at least 3.78 liters (1 gallon) shall be taken from each lot by the manufacturer and retained at their facility for possible future testing in accordance with the applicable procedure of ASTM D4057. This sample shall be retained by the manufacturer for a period of 3 years and shall be stored in a sealed 3.78-liter (1-gallon) container and labeled with information identifying the lot.

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4.4 Tests.

4.4.1 Significant figures. For purposes of determining conformance with the specified limits in this specification, observed or calculated values shall be rounded to the nearest unit in the right-most digit used in expressing the specification limit in accordance with the rounding method of ASTM E29.

4.4.2 Compatibility. A 1:1 mixture of the hydraulic fluid and each reference hydraulic fluid shall be subjected to the oxidation, rust prevention (in the presence of salt water), foam, and emulsion tests specified in table I (see 3.3). Mixtures meeting the requirements specified in table I are considered compatible fluids. The reference hydraulic fluid(s) shall be fluid(s) from the Qualified Products Database (QPD) associated with this specification. For contact information for obtaining reference hydraulic fluid(s), see 6.7.

4.4.2.1 Compatibility testing. Unless otherwise approved by NAVSEA, compatibility testing shall be conducted on all grades. If the manufacturer can verify in writing to NAVSEA that each grade of hydraulic fluid being qualified contains the same additive package, NAVSEA may grant reduced testing.

4.4.3 Homogeneity. After the pour point of the hydraulic fluid is determined, the test jar and hydraulic fluid shall be returned to the cooling bath, cooled to 12 °C (53 °F) below the pour point, and held at that temperature for 3 hours. The test jar shall be removed from the cooling bath and allowed to warm to room temperature. The test hydraulic fluid shall be visually examined for sedimentation or separation of insoluble materials. Then, the test jar and hydraulic fluid shall be placed in a heating bath, and the temperature of the bath shall be brought to 130±2.8 °C (266±5.0 °F) and held at that temperature for 1 hour. The test jar of hydraulic fluid shall be removed from the bath and allowed to cool in air to room temperature. The test hydraulic fluid shall be examined visually for sedimentation or separation of insoluble materials (see 3.4).

4.4.4 Appearance. The appearance of the hydraulic fluid shall be assessed by filling a clear quart-sized bottle with the hydraulic fluid and holding the sample in front of a strong light source. The hydraulic fluid shall be inspected for any cloudiness or haziness (see 3.5).

4.4.5 Toxicity and prohibited materials.

4.4.5.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.7.1 as required by the qualifying activity. The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the hydraulic fluid using data provided by the manufacturer/distributor to the NMCFHPC (see 3.7.1 and 6.6).

4.4.5.2 Prohibited materials. Prohibited materials shall be verified for conformance to 3.7.2 as required by the qualifying activity.

4.4.6 Off-gassing. The hydraulic fluid shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010, chapter titled “Material Control Program” (see 3.8 and 6.5). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA approved test facility (see 3.8). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.8).

4.4.7 2,6-Di-tert-butylphenol (DTBP). Analytical data from a laboratory acceptable to NAVSEA (see 6.3) shall be submitted to document the relative absence of DTBP, and if detected, shall not exceed 10 parts per million (see 3.9).

4.4.8 Chemical and physical characteristics. The hydraulic fluid chemical and physical requirements of 3.6 shall be verified in accordance with the test methods specified in table I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hydraulic fluid covered by this specification is intended for use in shipboard hydraulic systems and other applications where a high-grade hydraulic fluid having anti-corrosion and anti-oxidation properties without zinc additives or fire-resistant properties is required. Current stock of lubricating oil may be used until depleted; future requisitions for lubricating oil should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. Military symbol required (see 1.2).

d. Packaging requirements (see 5.1).

e. Activities requiring copies of completed Safety Data Sheets (SDS) (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 17672 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the QPD at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Copies of SD-6, “Provisions Governing Qualification,” are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil.

6.4 SDSs. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313 as specified in 6.2. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.5 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine’s atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled “Material Control Program”. Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.6 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC’s website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.7 Compatibility testing. Representative samples for compatibility testing are provided on request and are shipped to a laboratory acceptable to NAVSEA or contractor at Government expense. Information pertaining to

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obtaining representative samples may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil.

6.8 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, DoD Shelf-Life Management Program: Program Administration; and Volume 2, DoD Shelf-Life Management Program: Materiel Quality Control Storage Standards. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.9 International standardization agreement implementation. This specification implements North Atlantic Treaty Organization Standardization Agreement (NATO STANAG) 4601, Guide Specification for Hydraulic Fluids for Naval Applications (H-573, H-574 and H-576). When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.10 Subject term (key word) listing.

Anti-corrosion
Anti-oxidation

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – SH
Air Force – 68

Review activities:
Army – AT, CR4
Navy – AS
DLA – GS

Preparing activity:
Navy – SH
(Project 9150-2024-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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